

Convolutional Dynamically Convergent Differential Neural Network for Brain Signal Classification

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Abstract—The brain signal classification is the basis for the implementation of brain-computer interfaces (BCIs). However, most existing brain signal classification methods are based on signal processing technology, which require a significant amount of manual intervention, such as channel selection and dimensionality reduction, and often struggle to achieve satisfactory classification accuracy. To achieve high classification accuracy and as little manual intervention as possible, a convolutional dynamically convergent differential neural network (ConvDCDNN) is proposed for solving the electroencephalography (EEG) signal classification problem. First, a single-layer convolutional neural network is used to replace the preprocessing steps in previous work. Then, focal loss is used to overcome the imbalance in the dataset. After that, a novel automatic

dynamic convergence learning (ADCL) algorithm is proposed and proved for training neural networks. Experimental results on the BCI Competition 2003, BCI Competition III A, and BCI Competition III B datasets demonstrate that the proposed ConvDCDNN framework achieved state-of-the-art performance with accuracies of 100%, 99%, and 98%, respectively. In addition, the proposed algorithm exhibits a higher information transfer rate (ITR) compared with current algorithms.

Index Terms—Brain-computer interface (BCI), electroencephalography (EEG), machine learning, neural dynamic learning, neural network.

I. INTRODUCTION

HOW to better facilitate communication with individuals suffering from neurological or muscular disabilities is a noteworthy concern. In the past decades, researchers have proposed a variety of methods to address this problem, among which the brain-computer interface (BCI) offers a viable solution based on electrical brain signals, which can significantly improve the quality of life for patients [1].

For BCI systems, people tend to prefer noninvasive electroencephalography (EEG) as input signals, as it eliminates the need for additional surgical risks and has a high temporal resolution. P300 is an event-related potential, which is highly stable and easily detectable in EEG [2]. The P300 signal typically emerges around 300 ms after a visual stimulus, often manifesting as a positive potential waveform in the parietal and central regions of the EEG. Researchers arrange characters in the form of a matrix and stimulate P300 signals through random flashing of rows or columns, thus creating a character speller commonly referred to as the P300 speller [3]. To better evaluate the performance of P300 detection algorithms, many benchmark datasets, such as BCI Competition 2003 [4] and BCI Competition III [5], have been proposed.

The detection of P300 signals presents numerous challenges, such as low signal-to-noise ratio (SNR) and high dimensionality, which requires the classifier to extract useful features from a large amount of noise. Furthermore, the P300 signal exhibits significant variability among different subjects. Despite researchers using self-supervised learning [6] or transfer learning [7], [8], collecting data from each subject is typically necessary for training or fine-tuning the corresponding model. Since patient samples are limited, and the recording of EEG signals is a time-consuming process, very limited data can be obtained from subjects. In addition, there are far more negative samples than positive samples in P300 signals. This

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can potentially lead to a bias toward the negative samples, which can weaken the performance of the classifier.

In previous works, researchers manually removed data noise, selected channels, reduced dimensions, and ultimately classified P300 signals using traditional machine learning algorithms. Blankertz and colleagues [9] used the algorithm of K -nearest neighbors for the categorization of EEG signals. However, the outcomes of this algorithm's classification exhibited a lack of precision, and an issue of dimensional explosion was evident. Kaper et al. [10] used support vector machines (SVMs) to classify EEG signals after simple preprocessing and achieved comparable results. Solhjoo et al. [11] used a concealed Markov model to categorize EEG signals. However, this approach depended on temporal data, whereas collectors of data typically rearranged the set of information. As the top-ranking participant in the third BCI competition, Rakotomamonjy and Guigue combined 17 SVMs, referred to as the ensemble SVM (ESVM), achieving a 96.5% accuracy in recognizing characters [12]. Salvaris and colleagues [13] attained slightly inferior outcomes compared with the champions through the amalgamation of Fisher's linear discriminant (FLD) with discrete-wavelet transform (DWT). Nevertheless, this integration remains viable for the implementation in an adaptive online system. Despite achieving acceptable results on the dataset, cumbersome preprocessing operations and the weak fitting ability hindered further improvements in accuracy.

In recent times, due to the rise in GPU capabilities, deep learning has progressively outperformed conventional machine learning algorithms across various domains, including computer vision and the natural language processing [14]. Deep learning has also demonstrated great potential for P300 signal classification. Cecotti and Graser [15] introduced an EEG classification framework consisting of a convolutional neural network with four layers (referred to as CNN-1) and a composite model named MCNN-1, which is an ensemble of five CNN-1 models. This approach was designed to perform EEG classification tasks without the need for channel selection and preprocessing. Liu et al. [16] used batch normalization to mitigate overfitting in convolutional neural networks. Shan et al. [17] achieved remarkably high accuracy using only a single convolutional layer, called OLCNN, offering a new perspective to researchers. Kundu and Ari [7] proposed a method called MsCNN-TL-ESVM, which used 1-D convolution for channel selection, followed by multiscale convolutions to extract features, ultimately using SVM for classification. To enhance the precision of P300 identification, Wang et al. [8] made notable advancements by implementing capsule networks alongside spatiotemporal attention mechanisms, called ST-CapsNet. These studies have confirmed the effectiveness of deep learning in P300 signal detection tasks. However, to avoid problems such as overfitting and gradient vanishing, researchers still grapple with hyperparameter tuning when using deep learning algorithms. As mentioned earlier, EEG datasets typically have very limited sample sizes, which restricts the depth of the models. Researchers need to consider how to train more efficient and accurate models with the samples available. In addition, Dağ et al. [18] propose a novel

BCI classifier, called P3CNET. Cherloo et al. [19] treated P300 detection as a multiclassification task based on the relevant time of occurrence with stimuli. Liu et al. [20] use explainable AI to lightweight convolutional neural networks, i.e., P300MCNN. These methods have achieved promising results and are worth noting.

When training a P300 classifier, a severe imbalance in the number of positive and negative samples can lead to a bias toward the class with more samples. Some researchers used resampling techniques to balance the number of positive and negative samples [16]. However such a straightforward duplication of a minority of samples would elevate the risk of overfitting. Some researchers opted for ensemble methods to mitigate this issue [12]. However, this approach significantly increases the computational load, and real-time performance needs improvement. The focal loss, introduced by Lin et al. [21], was used to address the issue of imbalanced positive and negative samples in object detection. This augmentation serves to intensify attention toward challenging instances, contributing to the overall effectiveness of the method. Using focal loss allows us to eliminate the need for manually balancing the dataset or training multiple models, enabling us to correctly classify P300 signals with a single model.

In deep learning, people typically opted for gradient-based optimizers, such as gradient descent (GD), to train neural network models. Furthermore, there is another class of neural networks based on neural dynamic, characterized by exponential convergence properties, i.e., support vector neural network (SVNN) [22]. Applying neural dynamic methods can accelerate the training speed and enhance convergence accuracy. Zhang et al. [23] proposed an innovative dynamically convergent differential neural network (DCDNN) based on neural dynamic for EEG classification. This neural network provided better convergence while using a minimal number of parameters, thereby enhancing the performance of the EEG classifier. However, the derivation of this algorithm inevitably involved calculating the SVD decomposition of the entire dataset, with a high time complexity (i.e. $\mathcal{O}(mn^2)$ [24]). Besides, it is difficult to design more powerful and deeper networks due to the complexity of its weight updating algorithm.

In this article, a novel convolutional DCDNN (ConvDCDNN) is proposed, designed, and analyzed. Being different from previous work, the ConvDCDNN includes a single-layer convolutional neural network to directly process raw signals, uses focal loss to overcome the imbalance in the dataset, and proposed automatic dynamic convergence learning (ADCL) algorithm for training neural network.

The rest of this article is organized as follows. Section II introduces the datasets used in this article, along with the corresponding preprocessing procedures. Section III provides a detailed introduction to the neural network topology and the ADCL algorithm proposed in this article. Finally, in Section IV, we analyze the experimental results and compare them with those of previous algorithms.

The main contributions of this article are as follows.

- 1) Replacing traditional channel selection, feature extraction, and dimension reduction with a single-layer

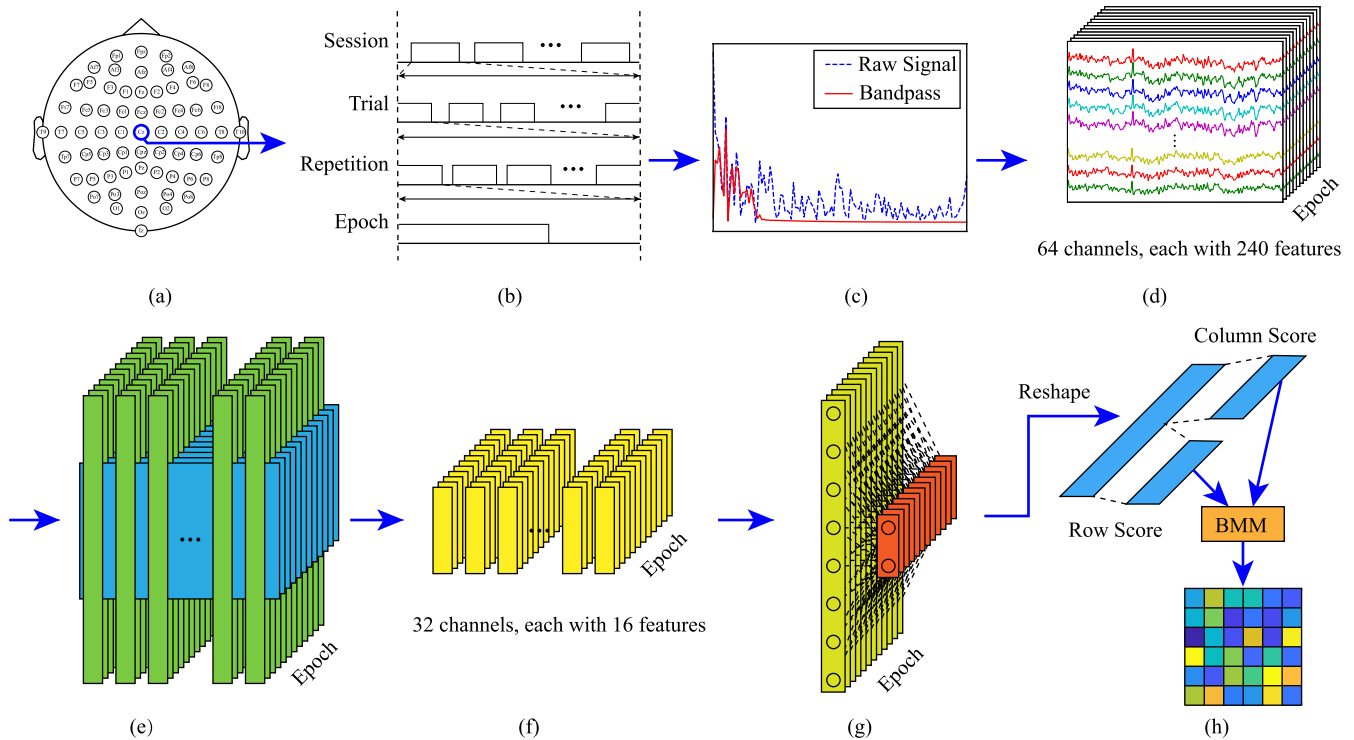


Fig. 1. Flowchart of the proposed method, BMM means batch matrix multiplication. (a) EEG cap. (b) Signal acquisition. (c) Filtering and normalization. (d) Sample of 12 epochs. (e) Convolutional layer. (f) Extracted feature. (g) Flatten and dense. (h) Character prediction.

convolutional neural network can effectively address the issues of feature loss and the reliance on cumbersome manual intervention.

- 2) Based on the combination of neural dynamic and automatic differentiation algorithms, the ADCL algorithm has been proposed for the first time in this article for training neural networks, offering faster convergence and higher accuracy.
- 3) Focal loss, early stopping, and dropout are used to address the issue of the imbalanced dataset, mitigating the problem of overfitting and enhancing the performance of the EEG classifier.

II. DATASET

The ConvDCDNN algorithm was evaluated using the publicly accessible Iib dataset from BCI Competition 2003 and the II dataset from BCI Competition III. Comparative analyses were conducted with several established algorithms using these datasets. This section emphasizes the datasets and outlines the preprocessing procedures involved.

A. Dataset Description

P300 speller is a character input system based on EEG signals. Users need to wear a head-mounted device capable of capturing EEG signals and focus their attention on the characters displayed on the screen. The head-mounted device will collect the potentials from the electrode channels through an A/D chip at a sufficiently high frequency. (For the dataset used in this article, the sampling frequency is 240 Hz.) The EEG cap is shown in Fig. 1(a), and the distribution of electrode positions is shown in Fig. 2.

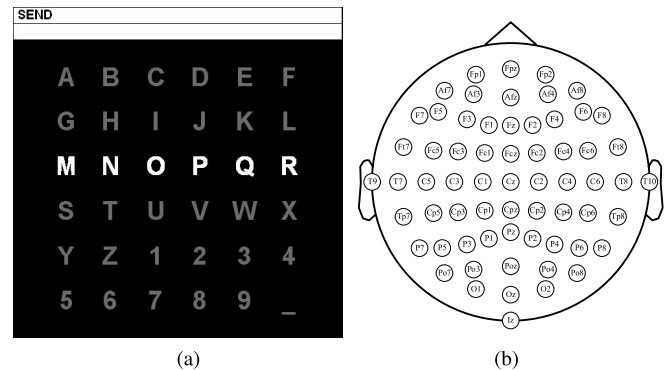


Fig. 2. P300 speller paradigm. (a) Screen. (b) 64 electrode channels.

Dataset Iib from BCI competition 2003 contains EEG signals of spelling 42 training characters and 31 testing characters. (There was a glitch in capturing the last three characters, so it was actually 39 training characters.) Similarly, dataset II from BCI Competition III includes two subjects. Each subject recorded 85 characters for training and 100 characters for testing.

Typically, characters are organized into a 6×6 matrix. After the user is prepared, 12 rows and columns on the screen start to flash successively, with each flash consisting of a 100-ms visual stimulus and a 75-ms blank period. A single flash is referred to as an epoch, and 12 epochs make up one repetition. Users need to go through 15 repetitions to complete one trial, to predict a character. In fact, users may not necessarily need to complete all the repetitions. For well-performing classifiers, typically, only the first few repetitions are often sufficient to predict the correct character.

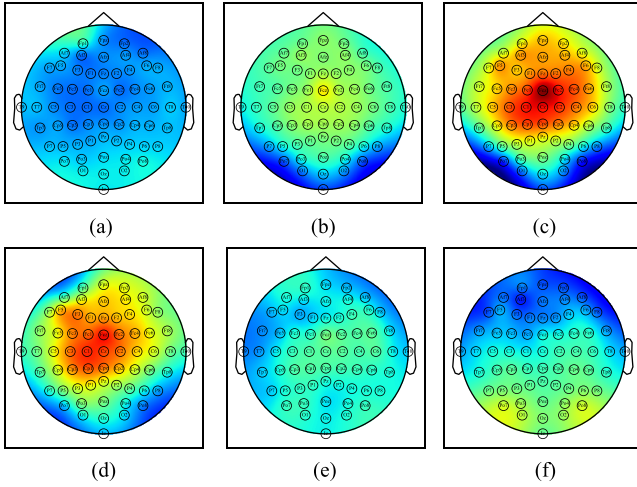


Fig. 3. Averaged topography with the target minus the nontarget component after visual stimulation. (a) 200 ms. (b) 250 ms. (c) 300 ms. (d) 350 ms. (e) 400 ms. (f) 450 ms.

B. Data Preprocessing

The P300 signal typically occurs between 100 and 300 ms after a visual stimulus [25], which can be seen intuitively from Fig. 3. As shown in Fig. 1(b), we use a 1-s time window starting at the onset of the visual stimulus to capture the P300 signal. The EEG signal typically has a low SNR, and attempting classification tasks without proper preprocessing often leads to low accuracy. We referred to a similar work and applied an eighth-order Chebyshev Type I bandpass filter to raw data, filtering it from 0.1 to 20 Hz [12]. The filtering result is shown as in Fig. 1(c). Subsequently, we use channelwise z -score standardization to ensure that the main part of the data is distributed between -3 and 3 , i.e.,

$$z = \frac{x - \bar{x}}{S} \quad (1)$$

where $\bar{x}$ represents the channel's mean, and S is the channel's standard deviation.

The significant difference from other works lies in dimensionality reduction and channel selection. Typically, researchers use various methods in the time or frequency domain to significantly reduce the number of features to avoid overfitting [26], [27]. However, this inevitably leads to a significant loss of information. In contrast, our preprocessing operations only consist of convolution and z -score standardization. We attempt to use a single-layer convolutional neural network to optimize the output features, preserving as much information as possible for the classification of brain signals.

After preprocessing, we randomly select 10% of the training samples as the validation set (need to keep the proportion of positive and negative samples) to evaluate the performance of the model during the training process. Eventually, we obtained the train, validation, and test datasets, as shown in Table I.

III. ConvDCDNN FRAMEWORK

In this section, the proposed ConvDCDNN framework is demonstrated, including the network topology and the ADCL algorithm based on neural dynamic.¹

¹The code is available at <https://github.com/BIRLab/ConvDCDNN>.

TABLE I
NUMBER OF P300s/NON-P300s FOR EACH DATASET

Dataset	Train		Validation		Test	
	P	N	P	N	P	N
BCI 2003	1053	5265	117	585	930	4650
BCI III A	2295	11475	255	1275	3000	15000
BCI III B	2295	11475	255	1275	3000	15000

TABLE II
ARCHITECTURE OF P300 CLASSIFIER IN ConvDCDNN

Layer	Filters	Kernel Size	Output	Params
Conv1D	32	15	(32, 16)	30752
Dropout	-	-	(32, 16)	-
Tanh	-	-	(32, 16)	-
Flatten	-	-	(512)	-
Dense	-	-	(2)	1026
Softmax	-	-	(2)	-

A. Neural Network Topology

The topology of ConvDCDNN consists of two components: the P300 classifier and the character classifier. All the learnable parameters of the P300 classifier are located in the convolutional layer and fully connected layer, and the neural network's structure is shown in Table II. The input to the P300 classifier is a $(N_{\text{epoch}}, N_{\text{channel}}, N_{\text{sample}})$ tensor shown in Fig. 1.

Next, we use the P300 classifier to predict whether each epoch contains a P300 signal. Note that all the P300 classifiers share the same parameters, which allows us to predict all the input signals in parallel. Throughout the entire prediction process, several layers of operations are required. We will now introduce each of these layers as follows.

- 1) *Conv1D*: The original EEG signal consists of 64 channels, each with 240 features. After convolution, the output contains 32 channels, each with only 16 features. Therefore, we believe that a single convolutional layer implicitly completes feature extraction, channel selection, and dimensionality reduction, thereby replacing traditional manual methods.
- 2) *Dropout*: Similar to many algorithms that use bagging to prevent overfitting, dropout serves as an implicit form of bagging, helping to mitigate overfitting in the model.
- 3) *Tanh*: Because our model has a low number of layers and the inputs are standardized using z -scores. A hyperbolic tangent (Tanh) function is a more suitable choice, because its derivative's range aligns with the distribution of z -scores.
- 4) *Flatten and Dense*: These two layers perform linear projections on the EEG features extracted by the convolution layer, resulting in the desired output dimensions.
- 5) *Softmax*: Softmax is a common output layer for classifiers, used to transform the classifier's output into probabilities ranging from 0 to 1.

Each character corresponds to $N_{\text{repetition}} \times N_{\text{epoch}}$ flashes, where $N_{\text{repetition}} = 1, 2, \dots, 15$. After passing through the P300 classifier and resizing (only use the probability of P300),

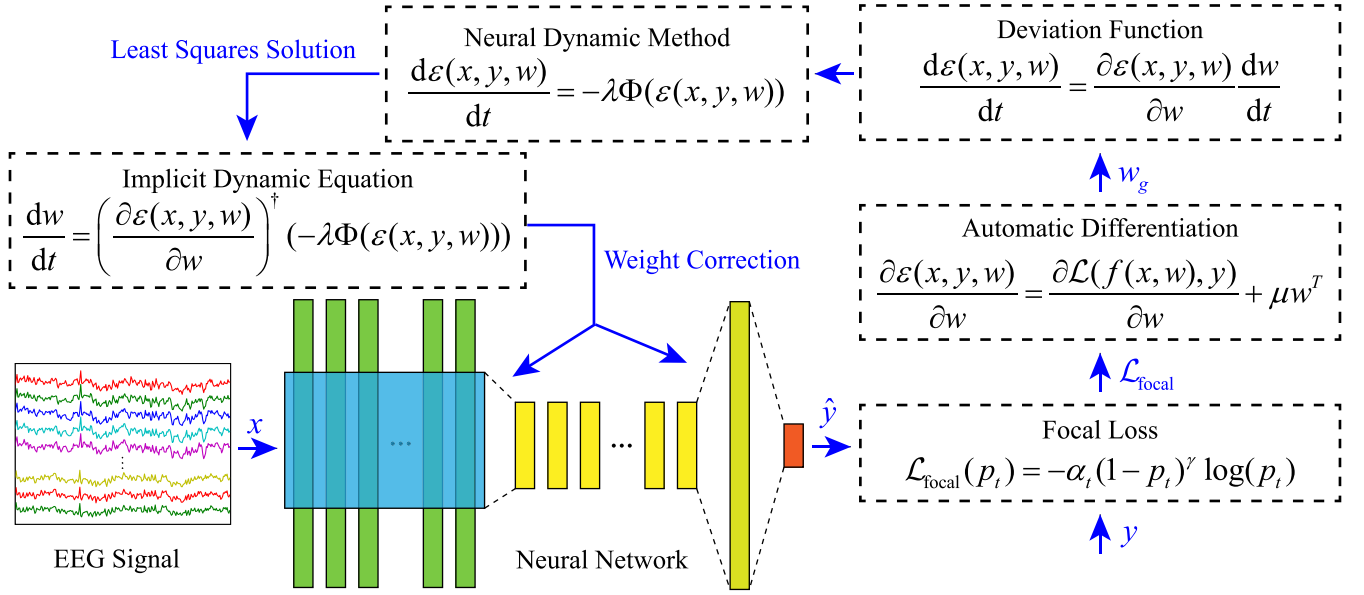


Fig. 4. Schematic of the ADCL algorithm.

the output size is $(N_{\text{repetition}}, N_{\text{epoch}})$. To obtain a probabilistic prediction $p(\text{char}_{ij})$ of the character, the product of the row and column probabilities needs to be calculated, i.e.,

$$\begin{aligned} p(\text{char}_{ij}) &= p(\text{row}_i \cap \text{column}_j) \\ &= p(\text{row}_i)p(\text{column}_j) \end{aligned} \quad (2)$$

where $p(\text{row}_i)$, $p(\text{column}_j)$ denotes the probability of whether the i th row, j th column contains P300 signals, respectively, $i, j = 1, 2, \dots, 6$. So, we split the P300 prediction results into rows and columns and perform a batch matrix multiplication operation to obtain probability predictions for 36 characters. The entire calculation process is illustrated in Fig. 1(h). Finally, we compute the average along the repetition dimension to obtain the final character prediction result.

B. Focal Loss

Classification tasks usually use the standard cross-entropy (CE) loss function. However, for datasets with extremely imbalanced sample sizes, the CE loss function may lead to overfitting. Researchers also mitigate overfitting by multiplying the CE loss with a weight coefficient, known as balanced CE (BCE). However, the results are still not satisfactory. To address this issue, we use focal loss instead of CE loss function to train our neural network model. The focal loss $\mathcal{L}_{\text{focal}}(p_t)$ is formulated as

$$\mathcal{L}_{\text{focal}}(p_t) = -\alpha_t(1-p_t)^\gamma \log(p_t) \quad (3)$$

where α_t is a weighting factor, $\gamma \geq 0$ is a tunable focusing parameter, and p_t refers to the probability prediction, defined as

$$p_t = \begin{cases} \hat{y}, & \text{if } \hat{y} \text{ is the probability of } y \\ 1 - \hat{y}, & \text{otherwise} \end{cases} \quad (4)$$

where y is the ground-truth label and $\hat{y}$ is the probability predicted by the neural network. In this article, α_t is determined based on the inverse class frequency, and γ is set to 2.

C. ADCL algorithm

The schematic of the ADCL algorithm is shown in Fig. 4. Suppose a neural network f takes input x and outputs $\hat{y}$, i.e.,

$$\hat{y} = f(x, w) \quad (5)$$

where f needs to be first-order derivable.

The solution of a neural network is typically formed as a single-objective optimization problem, and the criterion function is defined as

$$\mathcal{L}(\hat{y}, y) \geq 0 \quad (6)$$

where y is the ground-truth label, $w \in \mathbb{R}^m$ represents a vector consisting of all the learnable parameters of the neural network, and $\mathcal{L}(x, w) \in \mathbb{R}$ can be some common criterion functions, such as mean squared error loss (mse) or CE loss. Specifically, we use the focal loss function shown in (3).

Next, we define the error function $\epsilon(x, y, w)$ as follows:

$$\epsilon(x, y, w) = \mathcal{L}(f(x, w), y) + \frac{\mu}{2} \|w\|^2 \quad (7)$$

where μ is the L2 regularization factor, which can be used to avoid the overfitting problem.

Then formulate our optimization objective as follows:

$$w^* = \underset{w}{\operatorname{argmin}} \epsilon(x, y, w). \quad (8)$$

Consider the neural dynamic equation formulated as

$$\frac{dw}{dt} = -\lambda \Phi(\epsilon(x, y, w)) \quad (9)$$

where the activation function $\Phi(\cdot)$ needs to satisfy the following conditions:

$$\begin{cases} \Phi(x) = 0, & x = 0 \\ \Phi(x) > 0, & x > 0. \end{cases} \quad (10)$$

While using nonlinear activation functions may reduce the number of iterations required for convergence, it introduces

additional hyperparameters and uncertainties, making the computational process of training more complex and parameter tuning more difficult. We aim to minimize the difficulty of application and support some resource-constrained embedded devices, so we have decided to use linear activation functions. Specifically, we use $\Phi(x) = x$ to reduce computational overhead.

Take the first derivative of the error function $\varepsilon(x, y, w)$, we have

$$\frac{d\varepsilon(x, y, w)}{dt} = \sum_{j=1}^m \frac{\partial \varepsilon(x, y, w)}{\partial w_j} \frac{dw_j}{dt} = \frac{\partial \varepsilon(x, y, w)}{\partial w} \frac{dw}{dt}. \quad (11)$$

Let w_g be the Jacobian matrix of $\varepsilon(x, y, w)$ with respect to the parameters w , i.e.,

$$w_g = \frac{\partial \varepsilon(x, y, w)}{\partial w} = \frac{\partial \mathcal{L}(f(x, w), y)}{\partial w} + \mu w^T. \quad (12)$$

To compute w_g , we use the reverse accumulation automatic differentiation algorithm. Assume that we have a directed computational graph of $\mathcal{L}$, the nodes of which consist of operators or learnable parameters. Each learnable parameter w_j has a set of successors S_j defined as $S_j \in \{w_k \mid w_k \text{ is successor of } w_j\}$. Since $S_k \subset S_j$ when w_k is a successor of w_j , we must be able to compute $\partial \mathcal{L}(f(x, w), y)/\partial w$ recursively as

$$\frac{\partial \mathcal{L}(f(x, w), y)}{\partial w_j} = \sum_{w_k \in S_j} \frac{\partial \mathcal{L}(f(x, w), y)}{\partial w_k} \frac{\partial w_k}{\partial w_j}. \quad (13)$$

After computing w_g , (11) can be rewritten as

$$\frac{d\varepsilon(x, y, w)}{dt} = w_g \frac{dw}{dt}. \quad (14)$$

This is an underdetermined equation, and it has infinite solutions. To let the parameters update more smoothly, akin to minimizing the overall system's kinetic energy $\|dw/dt\|^2$, we seek the solution with the minimum L2-norm, that is,

$$\frac{dw}{dt} = w_g^\dagger \frac{d\varepsilon(x, y, w)}{dt} = w_g^\dagger (-\lambda \Phi(\varepsilon(x, y, w))). \quad (15)$$

Since w_g is a row vector, the pseudoinverse is the transposed vector divided by its squared magnitude, that is,

$$w_g^\dagger = \frac{w_g^T}{w_g w_g^T}. \quad (16)$$

Because of the design (9), $\varepsilon(x, y, w(t))$ is a monotonically bounded function. Thus, the convergence of the neural network is guaranteed.

D. Discretization of ADCL algorithm

According to the neural dynamic formula

$$\dot{w}(t) = \left(\frac{\partial \varepsilon(x, y, w(t))}{\partial w(t)} \right)^\dagger (-\lambda \Phi(\varepsilon(x, y, w(t)))) \quad (17)$$

we can express (17) in integral form

$$w(t + \Delta t) = w(t) + \int_t^{t+\Delta t} \dot{w}(\tau) d\tau \quad (18)$$

where Δt denotes the integration step length. When $\Delta t \rightarrow 0$, we can approximate the integral as the area of a rectangle, i.e.,

$$w(t + \Delta t) \approx w(t) + \dot{w}(t) \Delta t. \quad (19)$$

Let the initial parameter of the neural network be $w_0 = w(t_0)$, and let $w_k = w(t_0 + k\Delta t)$, where $k = 1, 2, \dots, \infty$. Then we can express the above equation in a discrete form, that is,

$$w_{k+1} = w_k + \dot{w}_k \Delta t = w_k + \left(\frac{\partial \varepsilon_k}{\partial w_k} \right)^\dagger (-\lambda \Phi(\varepsilon_k)) \Delta t \quad (20)$$

where Δt is the time step, a very small fixed constant. The equation can be rearranged as

$$w_{k+1} = w_k - \lambda^* \Phi(\varepsilon_k) \left(\frac{\partial \varepsilon_k}{\partial w_k} \right)^\dagger \quad (21)$$

where $\lambda^* = \lambda \Delta t$ is a fixed time constant (or a hyperparameter), and $\Phi(\varepsilon_k)$ is an activation function related to the current error. According to (16), we can simplify the calculation of the pseudoinverse as

$$w_{k+1} = w_k - \lambda^* \Phi(\varepsilon_k) \left(\frac{\partial \varepsilon_k}{\partial w_k} \right)^T / \left\| \frac{\partial \varepsilon_k}{\partial w_k} \right\|^2. \quad (22)$$

We can take the equivalent learning rate α_k as

$$\alpha_k = \lambda^* \Phi(\varepsilon_k) / \left\| \frac{\partial \varepsilon_k}{\partial w_k} \right\|^2. \quad (23)$$

Then the recursive formula (22) can be simplified as

$$w_{k+1} = w_k - \alpha_k \left(\frac{\partial \varepsilon_k}{\partial w_k} \right)^T. \quad (24)$$

In the above formula, α_k can be considered equivalent to the learning rate in GD. In contrast, GD uses a fixed learning rate, whereas the proposed α_k adjusts adaptively according to the training process, thus achieving dynamic convergence of the error.

In our experimental process, we found that by applying a numerical clamp on α_k , it is possible to mitigate the issue of divergence caused by improper parameter settings. So actually we use the following equivalent learning rate:

$$\alpha_k \leftarrow \min\{\alpha_k, \alpha_{\max}\} \quad (25)$$

where $\alpha_{\max}$ should be set to a relatively large value. So far, we give the pseudocode shown in Algorithm 1.

In most cases, the accuracy of rectangular integration is sufficient. However, if higher computational accuracy is required, we can use trapezoidal integration instead of rectangular integration

$$w_{k+1} = w_k - \frac{1}{2} \left(\alpha_k \left(\frac{\partial \varepsilon_k}{\partial w_k} \right)^T + \alpha_{k-1} \left(\frac{\partial \varepsilon_{k-1}}{\partial w_{k-1}} \right)^T \right). \quad (26)$$

IV. EXPERIMENTAL VERIFICATION

We conducted experiments on the BCI Competition 2003 and BCI Competition III datasets to compare the proposed method with the state-of-the-art algorithms. We also provide a detailed explanatory analysis of the proposed method, which helps further understand how it works.

Algorithm 1 Discrete ADCL Algorithm

Require: λ^* : Time constant
Require: μ : L2 regularization factor
Require: p : Patience
Require: n : Batch size
Initialize parameters w of the neural network
Initialize optimal parameters $w^* \leftarrow w$
Initialize $F_1^* \leftarrow 0$
while F_1^* has improved in any of the past p steps. **do**
Randomly take n samples from the dataset $\mathcal{D}$ as x
Forward propagation $\hat{y} \leftarrow f(x, w)$
Calculate $\mathcal{L}_{\text{focal}}(\hat{y}, y)$ according to (3)
Calculate $\partial \varepsilon(x, y, w) / \partial w$ according to (12)
Update α according to (23) and (25)
Update the parameters $w \leftarrow w - \alpha(\partial \varepsilon / \partial w)^T$
Calculate the validation F_1 score.
if $F_1 > F_1^*$ **then**
Update optimal parameters $w^* \leftarrow w$
Update $F_1^* \leftarrow F_1$
end if
end while
return Optimal parameters w^*

TABLE III
HYPERPARAMETER SETTINGS FOR LEARNABLE PARAMETERS

Parameter	λ^*	$\alpha_{\max}$	μ
w_k^c	1e-2	1e-2	1e-5
w_{bias}^c	1e-5	1e-2	1e-5
w_{proj}^d	1e-3	1e-2	1e-5
w_{bias}^d	1e-4	1e-2	1e-5

A. Experimental Setup

Due to the imbalance of positive and negative samples in the P300 dataset, evaluating samples based on accuracy and loss is unreliable. F_1 score is the harmonic mean of precision and recall, and therefore, it will not be affected by the imbalance of positive and negative samples. So we use the F_1 score to evaluate the model in the validation set

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}}. \quad (27)$$

The definitions of precision and recall are as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}; \quad \text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (28)$$

where TP, FP, and FN represent the numbers of true positives, false positives, and false negatives, respectively.

We use the same hyperparameters across different subjects, which are documented in Table III. Besides, we use early stopping to terminate training when the F_1 score on the validation dataset no longer improves, which helps avoid the overfitting problem. To ensure the reproducibility of our proposed algorithm, we fixed all the random number seeds during training. Necessary information for reproduction is presented in Table IV.

TABLE IV
NECESSARY INFORMATION REQUIRED FOR REPRODUCTION

Item	Value
CPU	Intel i9 14900K
GPU	NVIDIA RTX 4090
Framework	PyTorch 2.0
Batch Size	128
Dropout	0.25
Inference Time	<100 us

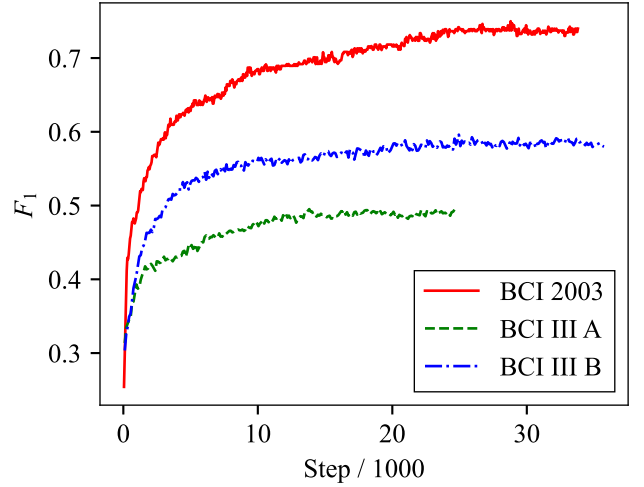


Fig. 5. F_1 score for single P300 detection on validation dataset during the training process.

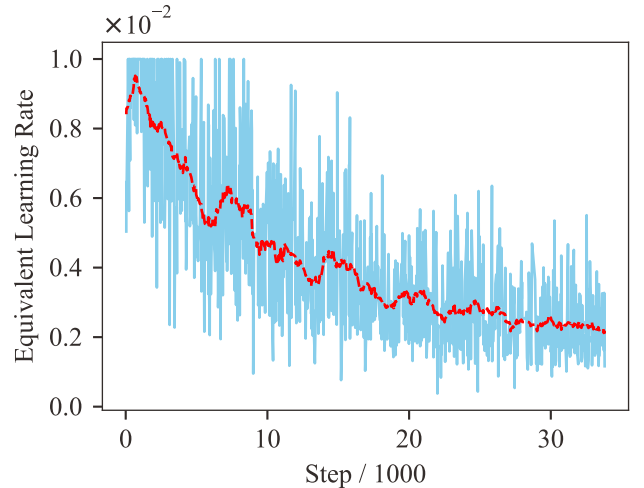


Fig. 6. Equivalent learning rate curve for w_k^c during the training process of the BCI 2003 dataset. The sky-blue solid line represents the equivalent learning rate, and the red dashed line represents the curve after average filtering.

B. Results

The F_1 score on the validation dataset throughout the training process is shown in Fig. 5. As can be seen from the figure, the proposed method converges quickly on different datasets and does not suffer from overfitting problems. We can also explain why the proposed method converges faster and has higher accuracy based on the equivalent learning rate curve as shown in Fig. 6. It can be intuitively seen that in

TABLE V
CHARACTER RECOGNITION ACCURACY OF DIFFERENT METHODS ON BCI COMPETITION 2003 AND BCI COMPETITION III

Datasets	Methods	Epoch														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
BCI Competition 2003	ConvDCDNN	83.9	96.8	100	100	100	100	100	100	100	100	100	100	100	100	100
	E-DCDNN [23]	77.4	83.9	90.3	100	100	100	100	100	100	100	100	100	100	100	100
	SVNN [22]	61.3	74.2	90.3	100	100	100	100	100	100	100	100	100	100	100	100
	BN3 [16]	77.4	74.2	87.1	90.3	93.5	96.8	96.8	96.8	100	100	100	100	100	100	100
	CNN-1 [15]	67.7	80.6	93.5	100	100	100	100	100	100	100	100	100	100	100	100
	MCNN-1 [15]	74.2	83.9	93.5	100	100	100	100	100	100	100	100	100	100	100	100
	MsCNN-TL-ESVM [7]	87.1	90.3	96.8	100	100	100	100	100	100	100	100	100	100	100	100
	ST-CapsNet [8]	87.1	93.5	96.8	100	100	100	100	100	100	100	100	100	100	100	100
	OCLNN [17]	74.2	83.9	90.3	100	100	100	100	100	100	100	100	100	100	100	100
	P300MCNN [20]	77.4	93.5	96.8	100	100	100	100	100	100	100	100	100	100	100	100
BCI Competition III A	ConvDCDNN	26	43	61	70	77	82	84	87	88	92	93	96	98	98	99
	E-DCDNN	14	38	56	58	65	72	76	82	86	88	87	93	97	97	98
	SVNN	17	36	61	63	72	76	78	80	82	89	93	93	97	97	99
	BN3	22	39	58	67	73	75	79	81	82	86	89	92	94	96	98
	CNN-1	16	33	47	52	61	65	77	78	85	86	90	91	91	93	97
	MCNN-1	18	31	50	54	61	68	76	76	79	82	89	92	91	93	97
	MsCNN-TL-ESVM	24	38	46	50	60	70	72	79	84	86	89	89	92	94	96
	ST-CapsNet	18	31	53	56	68	79	82	85	84	88	89	92	92	95	98
	OCLNN	23	39	56	63	73	79	82	85	90	91	94	95	95	96	99
	P300MCNN	23	30	52	60	67	75	79	83	84	87	89	96	94	93	98
BCI Competition III B	ConvDCDNN	48	65	73	76	84	86	88	91	94	95	96	96	95	96	98
	E-DCDNN	38	59	71	71	79	81	84	89	91	91	90	91	91	94	98
	SVNN	32	50	60	67	77	81	88	91	93	95	98	98	97	97	97
	BN3	47	59	70	73	76	82	84	91	94	95	95	95	94	94	95
	CNN-1	35	52	59	68	79	81	82	89	92	91	91	90	91	92	92
	MCNN-1	39	55	62	64	77	79	86	92	91	92	95	95	95	94	94
	MsCNN-TL-ESVM	40	59	67	74	79	84	90	92	94	97	96	98	97	97	96
	ST-CapsNet	41	61	66	78	85	86	92	90	91	95	96	96	95	97	96
	OCLNN	46	62	72	79	84	87	89	92	94	96	97	97	97	98	98
	P300MCNN	42	56	67	65	71	77	80	85	90	90	89	93	93	94	97

the early stages of training, the equivalent learning rate is relatively high, and the error decreases rapidly as a result. The equivalent learning rate gradually decreases throughout the training process, ultimately leading to higher convergence accuracy.

Subsequently, we calculated the accuracy of the proposed method on the test dataset and compared it with current methods [7], [8], [15], [16], [17], [20], [22], [23]. Specifically, the accuracy is calculated as the fraction of the correctly predicted characters among all the characters. The results of the comparison are shown in Table V. Our algorithm achieved state-of-the-art performance with accuracies of 100%, 99%, and 98% on the BCI Competition 2003, BCI Competition III A, and BCI Competition III B datasets, respectively, after a full 15 epochs.

For well-performing classifiers, typically, only the first few repetitions are often sufficient to predict the correct character. Considering that users always want to input characters as quickly as possible, we can measure the classifier's performance through the information transfer rate (ITR) as

shown below

$$\text{ITR} = \frac{60(P \log(P) + (1 - P) \log(\frac{1-P}{N-1}) + \log(N))}{T} \quad (29)$$

where P represents the accuracy of recognition, N stands for the total number of classes (in this context, N equals 36), and T signifies the time required for recognition, as specifically defined as

$$T = 2.5 + 2.1N, \quad N = 1, 2, \dots, 15. \quad (30)$$

To demonstrate the advantages of the proposed ConvDCDNN algorithm, we calculate ITR as (29) and compare it with other methods, as shown in Fig. 7. It can be observed that our algorithm outperforms most algorithms in 15 epochs. In the BCI Competition 2003 dataset, the character recognition accuracy of the proposed method is very high and maintains the maximum ITR after the second epoch. For the BCI Competition III dataset, the proposed method achieves the maximum value within the first three epochs and maintains a relatively higher value in the subsequent epochs.

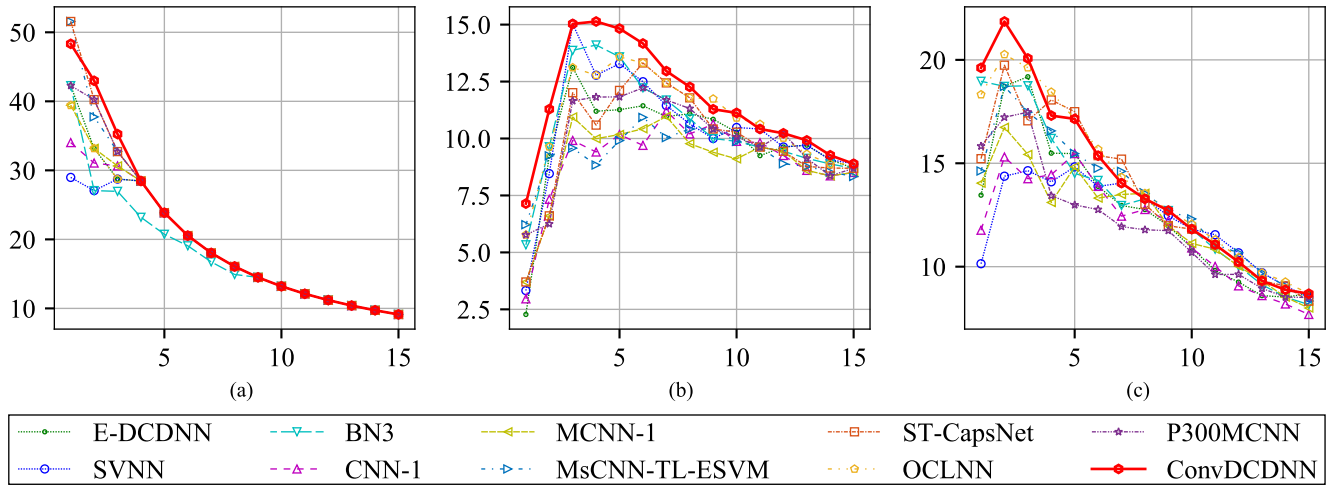


Fig. 7. ITR Comparison under 15 repetitions. Datasets (a) BCI 2003, (b) BCI III A, and (c) BCI III B.

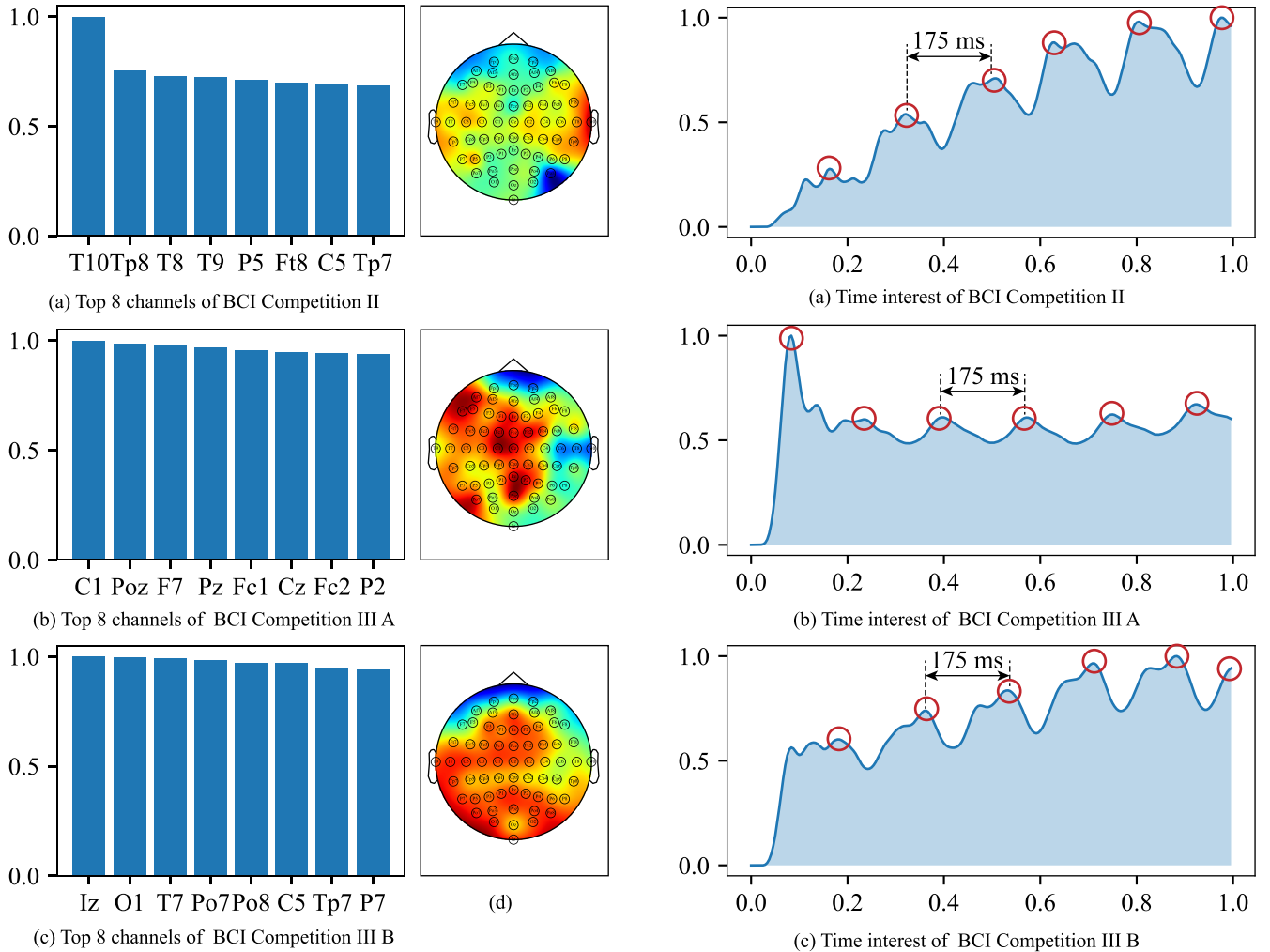


Fig. 8. (a)–(c) Top ten channels. (d) Heatmaps of interest corresponding to (a)–(c), respectively.

Fig. 9. (a)–(c) Interest of the proposed algorithm for different times. The x -coordinate indicates the time in seconds since the occurrence of the visual stimulus.

C. Discussion

1) *Implicitly Channel Selection*: We use the Grad-CAM [28] algorithm to analyze how the proposed model makes decisions and where its interest lies. Briefly, the

Grad-CAM algorithm computes the partial derivatives of the class score with respect to the activation of the k th feature layers and then performs global average pooling to obtain the

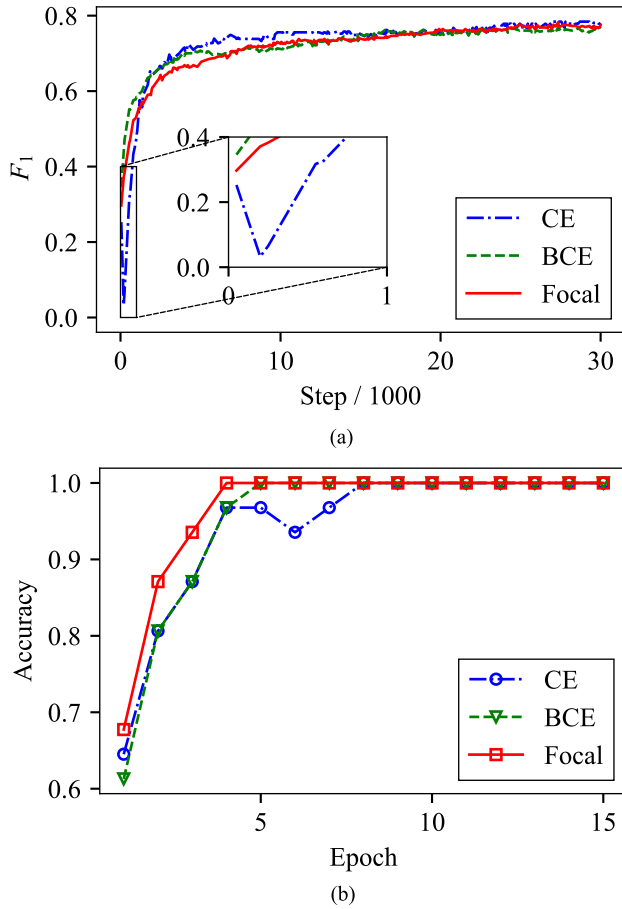


Fig. 10. Focal loss versus other loss functions. (a) Validation set F_1 curves using different loss functions. (b) Final character recognition accuracy using different loss functions.

weights for each feature layer

$$\alpha_k^c = \frac{1}{N} \sum_{i=1}^N \frac{\partial y^c}{\partial A_i^k} \quad (31)$$

where A^k denotes the activation value of the k th layer and y^c denotes the class score. After that, we need to compute the weighted average of these feature layers

$$\text{CAM} = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right) \quad (32)$$

where $\text{ReLU} = \max\{0, x\}$ is used to get the features that have a positive impact on the results.

Fig. 8 illustrates the top eight channels with the highest interest values for each subject, along with an intuitive heatmap. We can find that the interest of channels varies significantly from different subjects. However, the ConvDCDNN proposed in this article is able to cover this discrepancy by automatically selecting the appropriate channel during the training process, which proves that the design of single-layer convolution can replace the traditional channel selection and dimensionality reduction operations.

2) *Time Interest*: We analyze the interest of the proposed algorithm for different times as shown in Fig. 9. Wherever the P300 signal appears, our algorithm shows a high level of

interest (marked by red circles). This shows that the proposed algorithm is able to correctly recognize the features of the P300 signal. We also found something noteworthy in that our algorithm does not focus only on the 300 ms (the time at which the P300 signal theoretically appears) but on the entire time period. In addition, some distinct peaks appeared at intervals of approximately 175 ms, which just equaled the intervals of the visual stimuli (100 ms of intensification and 75 ms of blank). We hypothesize that this phenomenon results from the overlap of multiple visual stimuli in intercepted time window.² We believe that this overlap leads to a performance improvement because our algorithm can use more information to infer whether a P300 signal is present at 300 ms. For example, if P300 signals are detected at both 300 and 475 ms, with the signal at 475 ms being more significant, it could be inferred that the detection at 300 ms might be a result of interference leading to an erroneous detection.

3) *Impact of Focal Loss*: Additional experiments were used to demonstrate the improvements brought by focal loss. Under the hyperparameters, we train the neural network with CE, BCE, and focal loss, respectively. The results of the experiment are shown in Fig. 10. As can be seen from the figure, F_1 score of CE loss function decreases abruptly at the beginning of training, due to overfitting to negative samples. Although the three loss functions ultimately achieve similar F_1 scores, as shown in Fig. 10(b), the character prediction results of the focal loss function are significantly higher than those of the others. This illustrates that the bias still exists for negative samples in both CE and BCE losses, but focal loss works well to eliminate this bias.

V. CONCLUSION

To achieve high classification accuracy and as little manual intervention as possible, a ConvDCDNN has been proposed for solving the EEG signal classification problem. Compared with previous work, the proposed ConvDCDNN effectively reduces the workload associated with EEG data processing. In addition, the use of neural dynamic optimization algorithm has reduced training time and improved accuracy. Comparative experiments have been conducted on the BCI Competition 2003 and BCI Competition III datasets, and the experiments have verified that ConvDCDNN can achieve state-of-the-art accuracy (100%, 99%, and 98% at BCI Competition 2003, BCI Competition III A, and B, respectively) after completing 15 epochs in character classification, while maintaining high ITRs.

Future work will focus on further improving the ADCL algorithm by incorporating features such as heuristics and adaptiveness. In addition, we will attempt to apply the ConvDCDNN framework in practical scenarios beyond real-time BCI systems, such as disease diagnosis and emotion recognition.

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²The greater importance of the signal at 475 ms may be due to the deflection of the EEG signal and not because of a delayed P300.

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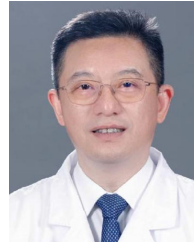


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